

Digital design verification

Cocotb verification for ALU

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Submitted to:

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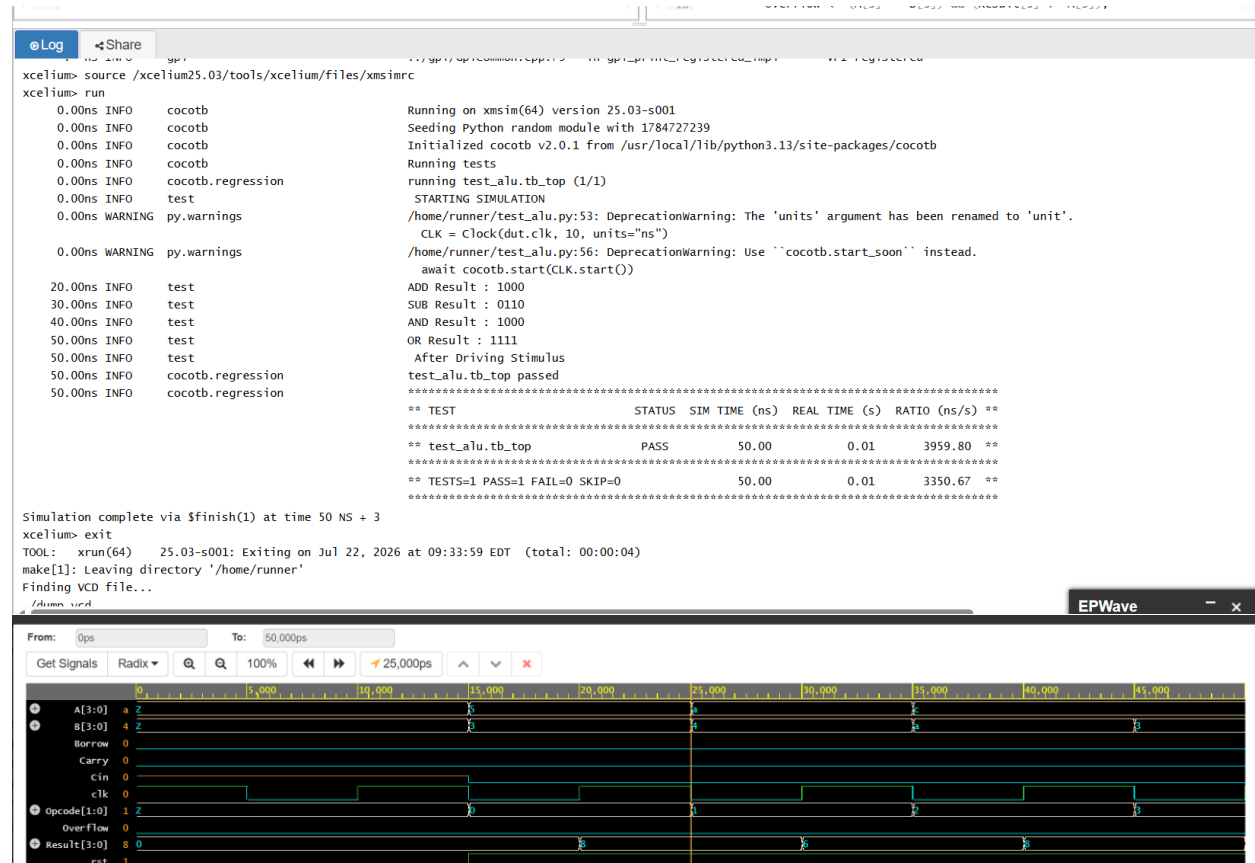
The flow of the verification method:

1. Simulator starts
2. Python environment initializes
3. tb_top() starts
4. Clock Starts Running
5. driving_stimulus
6. Apply active-low reset
7. Loop over operations
8. Check results
9. Test passes / finish

The code starts with importing the libraries of the cocotb needed, and start to put the testing scenarios as a form of driving stimulus. The block of cocotb.test is the one that initialize the program with addressing the clock and the start and start_soon functions.

The makefile role is to load Cocotb's library into the simulator so that Python can read and write hardware signals while the simulation is running.

The results:



The input data are detected at the posedge of the clock, and then the result comes out as illustrated in the waveform and in the log above.